



Workshop on the Use of Alternative Data Sources for Consumer Price Statistics

UN Regional Hub for Big Data and
Data Science in Brazil

UN Task Team on Scanner Data

22–26 June 2026

 **UN Big Data Learning**

2. ADS Acquisition

UN-CEBD Task Team
on Scanner Data

Outline

1. Acquiring Scanner Data
2. Scraping data from the web
3. Scraping data from the web
4. Administrative Data Acquisition
5. Conclusions



1. Acquiring Scanner Data for CPI

Contents

1. Developing a strategy to acquire scanner data
2. Scanner data requirements for Consumer price indices
3. Negotiation strategy & requirements for scanner data deliveries from large-scale trade retailers
4. Managing the acquisition of scanner data & skills require
5. IT structure and monitoring
6. Highlights



1. Developing a strategy to acquire scanner data

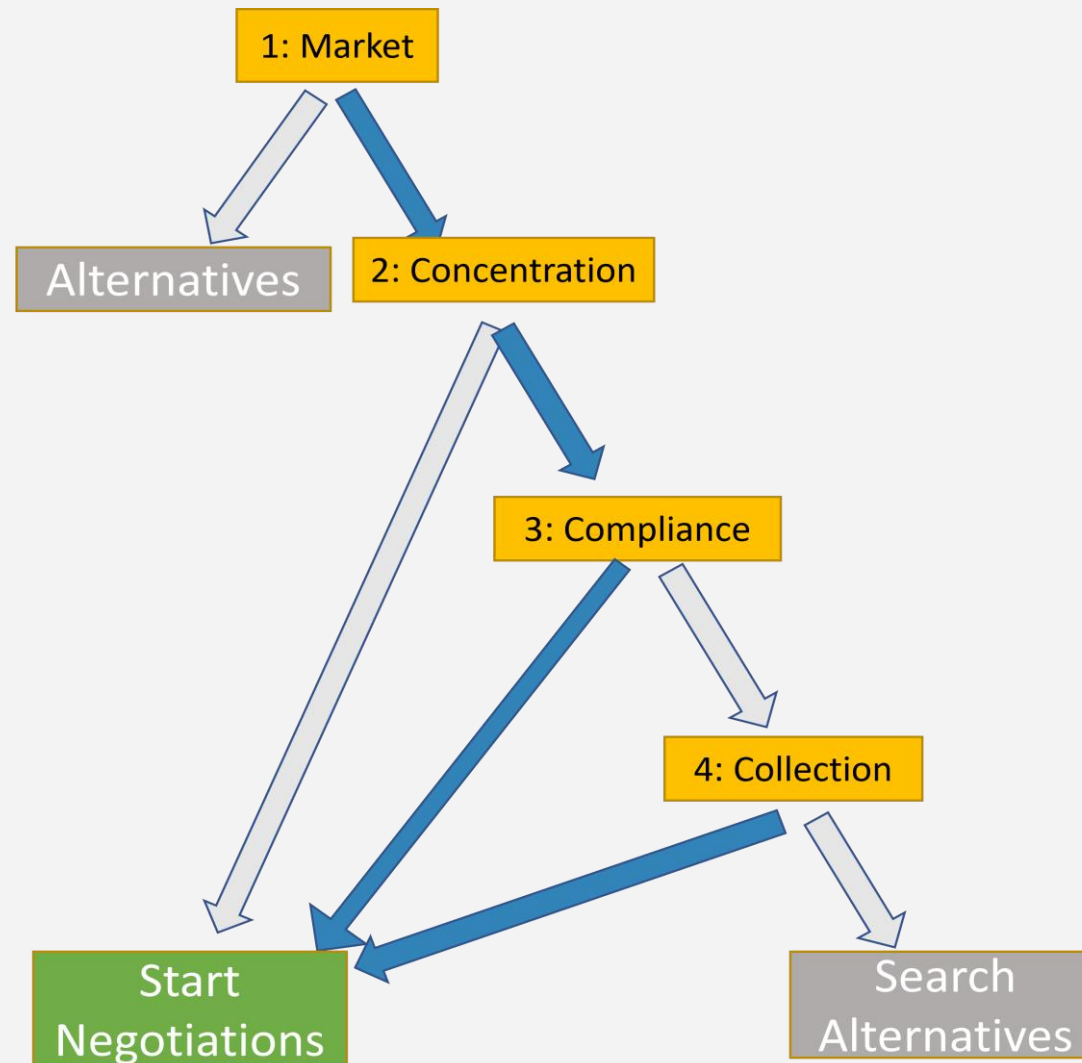


A decision tree to guide NSO choices

- This section describes step-by-step the pathway from initial contact with the data provider to the acquisition of scanner data
- A “decision tree” is presented
- The graphical representation of this decision tree will help us to follow the sequence of steps and nodes
- Please note that thresholds for decision are purely indicative and may need to be adjusted to your specific case



A decision tree to guide NSO choices



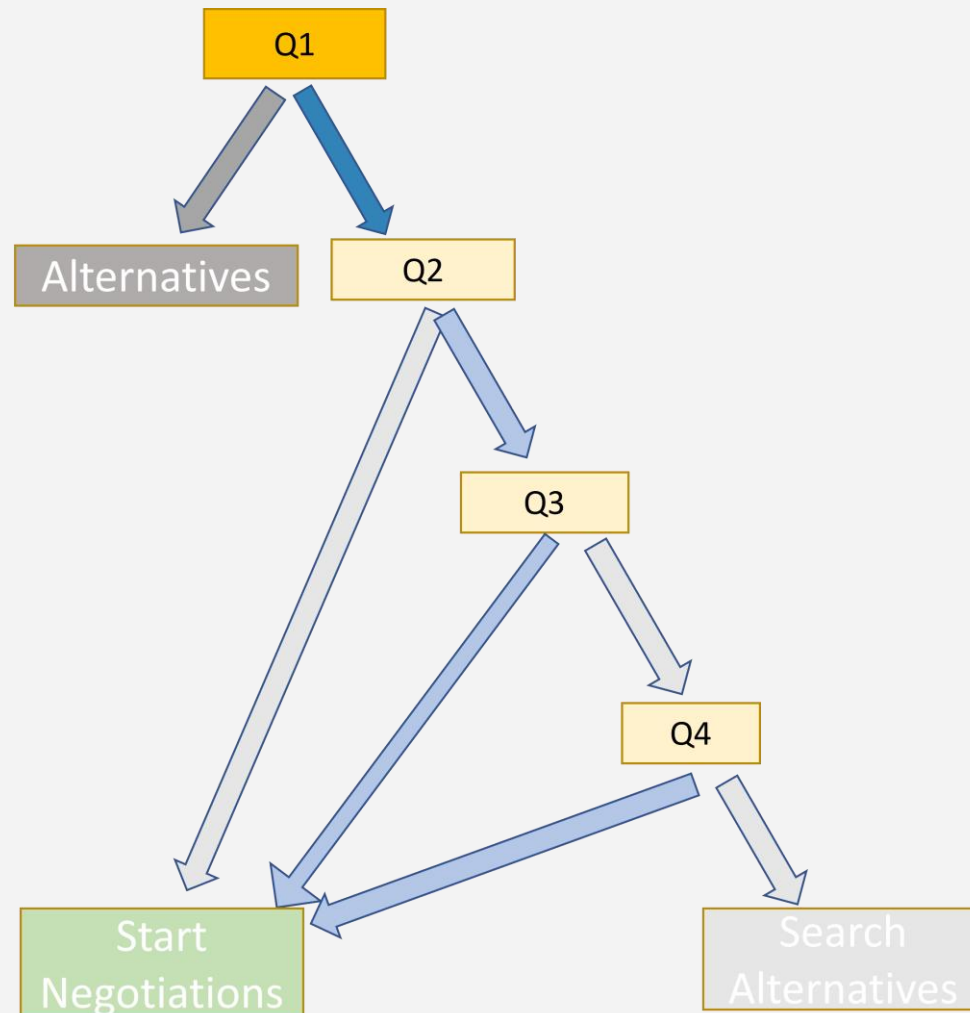
For this next section, we will work through the first 4 nodes of the “decision tree” in detail



First step/node: Analysis of local markets

First step/node:

- **Q1:** Is the large-scale retail trade distribution predominant or not?
- Analyze the question by group of products or by COICOP expenditure division
- For each group of products or COICOP division, detect the characteristics and estimate the weights of each retail trade distributional channel in terms of turnover (an example of a summary table is showed in the next slides and available in [GitHub area](#))



First step/node: Analysis of local markets

Table 1 (1). Example of analysis of the local retail trade market by group of products

GROUP OF PRODUCTS	Large scale retail trade channels	Traditional distribution	Service provider establishment, companies, institutions	E-commerce	Others	Total
Processed food including alcoholic beverages	60	20	0	20	0	100
Unprocessed food	40	40	0	20	0	100
Clothing and footwear	40	40	0	20	0	100
Household appliances	10	10	50	20	10	100
Furnishings, household equipment and routine household maintenance	40	30	10	20	0	100
Medicines and health products	20	30	40	10	0	100
Personal transportation equipment	10	30	30	30	0	100
Transportation services	30	10	20	40	0	100
Fuels	90	10	0	0	0	100
Communication equipment and consumer electronics	30	20	20	30	0	100
Recreation and culture	20	20	30	20	10	100
Travel	40	10	10	40	0	100
Education	0	0	100	0	0	100
Restaurants and hotels	30	20	10	40	0	100
Others	20	20	20	30	10	100



First step/node: Analysis of local markets

Table 1 (2). Example of analysis of the local retail trade market by COICOP 2018 division

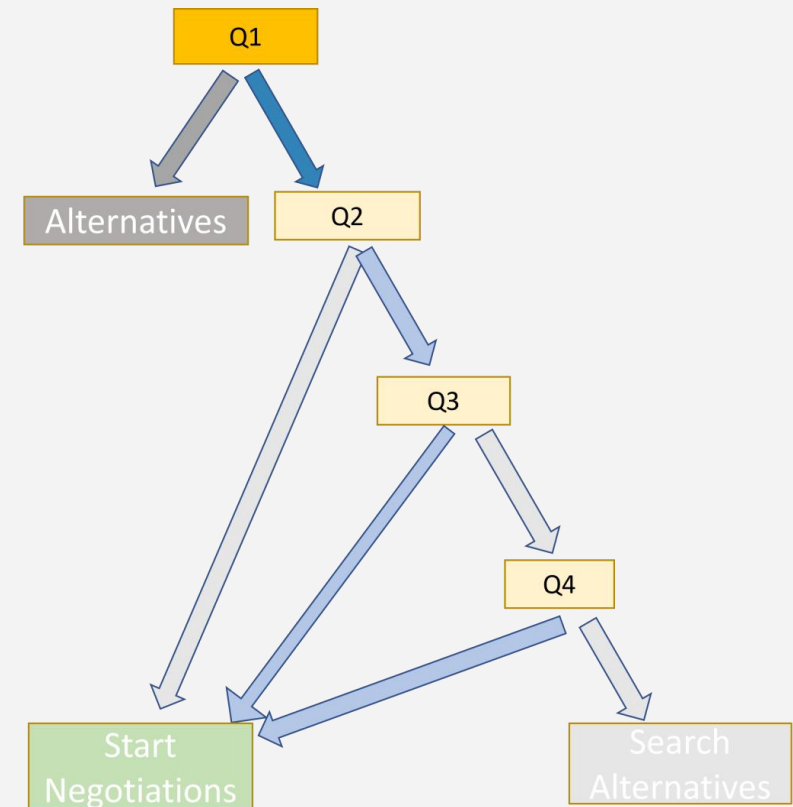
COICOP EXPENDITURE DIVISION	Large scale retail trade channels	Traditional distribution	Service provider establishment, companies, institutions	E-commerce	Others	Total
Food and non-alcoholic beverages	60	20	0	20	0	100
Alcoholic beverages, tobacco	50	30	0	15	5	100
Clothing and footwear	40	40	0	20	0	100
Housing, water, electricity, gas and other fuels	10	10	50	20	10	100
Furnishings, household equipment and routine household maintenance	40	30	10	20	0	100
Health	20	30	40	10	0	100
Transport	10	30	30	30	0	100
Information and communication	60	10	10	20	0	100
Recreation, sport and culture	20	20	30	20	10	100
Education services	0	0	100	0	0	100
Restaurants and accomodation services	0	0	60	40	0	100
Insurance and financial services	20	20	20	30	10	100
Personal care, social protection and miscellaneous goods and services	40	30	20	10	0	100



First step/node: Analysis of local markets

First step/node:

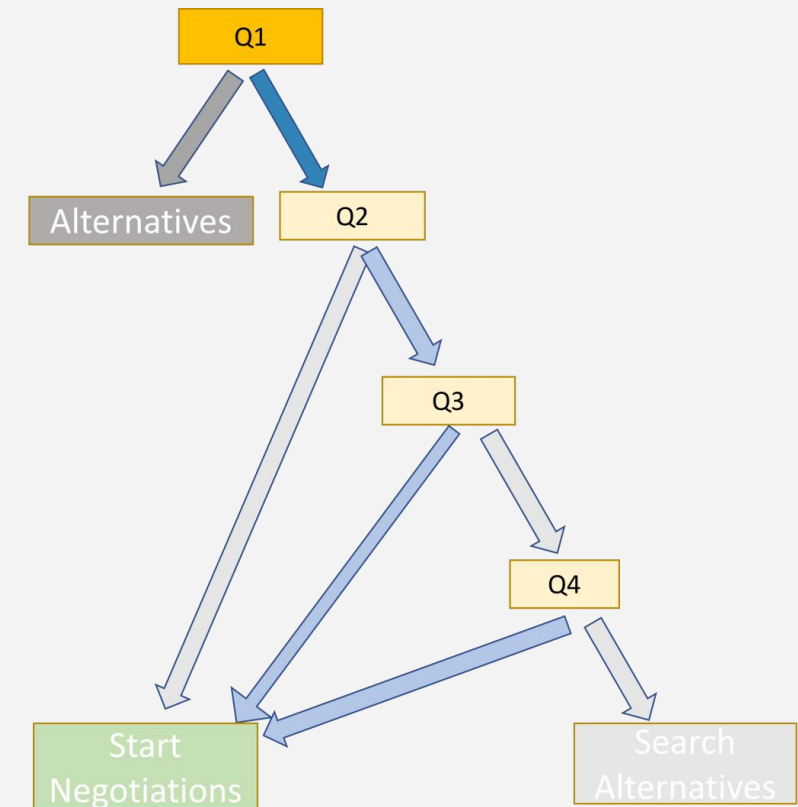
- **Q1:** Is the large-scale retail trade distribution predominant or not?
- **Analyze** the question by group of products or by COICOP expenditure division
- For each group of products or COICOP division, **detect** the characteristics and estimate the weights of each retail trade distributional channel in terms of turnover ([a summary table could be produced](#))
- **Decision:** if large-scale retail trade distribution
 - is deemed relevant in terms of turnover, [go to the 2nd decision node](#)
 - is not deemed relevant in terms of turnover, [consider alternatives](#)



Second step/node: Analysis of the level of concentration of the large-scale retail trade distributional channel

Second step/node:

- **Q2:** What is the level of concentration of the large-scale retail trade distributional channel, in the group of products or COICOP division being considered?
- Need to understand the market dominance (turnover percentages of companies in the large-scale retail trade market)
- **Decision:** if 70-80% of the turnover of large-scale retail trade distribution is represented by:
 - ✓ More than 4-5 companies – **refer to Case 1**
 - ✓ Less than or equal to 4-5 companies – **refer to Case 2**



Second step/node: Analysis of the level of concentration of the large-scale retail trade distributional channel

Case 1

- 70-80% of the turnover of large-scale retail trade distribution is represented by more than 4-5 companies
 - **Look** for centralized access to companies
 - **Negotiate** contacts with national branch of GS1 or with associations representing companies in each market



Second step/node: Analysis of the level of concentration of the large-scale retail trade distributional channel

Case 2:

- 70-80% of the turnover of large-scale retail trade distribution is represented by less than 4-5 companies
 - ✓ Promoting bilateral contacts and meetings
 - ✓ For example: in Australia, there are four grocery companies that represent 80% of retail trade. In this case, negotiations have occurred separately with each company rather than through a centralized channel.



Second step/node: Analysis of the level of concentration of the large-scale retail trade distributional channel

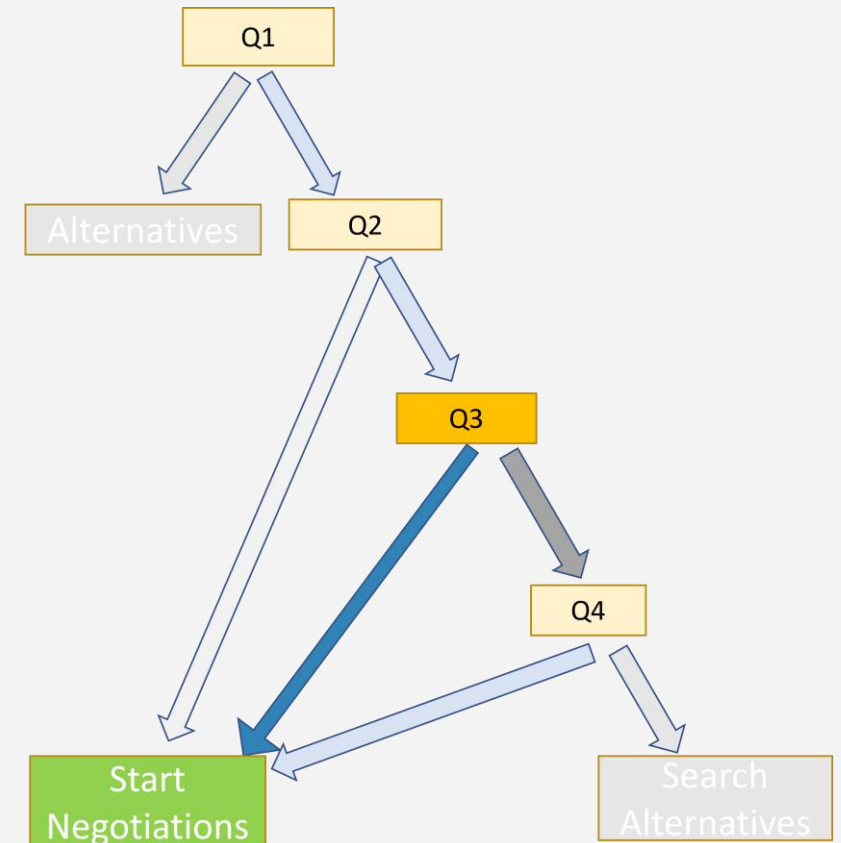
- The approach to the negotiations for scanner data with large-scale retail distribution in both cases 1 and 2 is aimed at:
 - ✓ Explaining why it is important for NSOs to receive scanner data
 - ✓ Stressing the benefits to the NSO in improving the quality of information for the measurement of price changes
 - ✓ Stressing the benefits to the company as the scanner data enables measurement of the company's contribution to price change



Third step/node: Agreement or reluctance of companies to provide scanner data to NSOs

Third step/node:

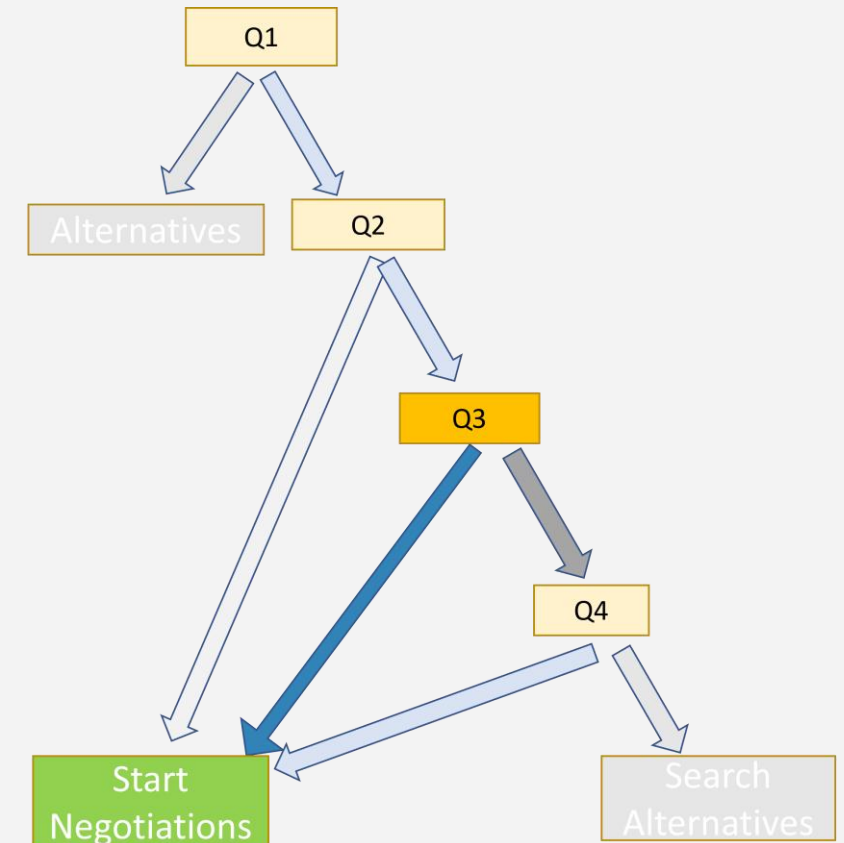
- **Q3:** Do the companies of large-scale retail trade distribution agree to provide scanner data to NSOs or are they reluctant?
- ✓ **Yes:** if they agree, start **negotiations** with:
 - GS1 or sector Association where the market is fragmented (Case 1) or
 - Companies in bilateral discussions if concentrated market (Case 2)
- ✓ **No:** if they are reluctant, go to the fourth step/node



Fourth step/node: reluctance of companies to provide scanner data to NSOs

Fourth step/node:

- **Q4:** Are the companies reluctant to share scanner data due to the burden?
- **Yes:** Explore the possibility to receive the data from market analysis companies that are already recipients of the data. If this is the case, set up a three-stepped approach
 - ✓ 1. NSO asks the market analysis company for the scanner data
 - ✓ 2. The large-scale retail trade company authorizes it
 - ✓ 3. The market analysis company provides the NSO with the requested data



Fourth step/node: reluctance of companies to provide scanner data to NSOs

Fourth step/node:

- **Q4:** Are the companies reluctant to share scanner data due to the burden?
- **No:** Explore other options for obtaining the scanner data:
 - ✓ Look at the legislation and appeal to law if available
 - ✓ Start bilateral discussions for approval to web scrape data (no longer scanner data!)



Next steps

- Let's focus on the case where the large-scale retail trade company or market analysis company agrees to deliver scanner data to the NSO
- What detail needs to be negotiated and agreed with the company?



2. Scanner data requirements for Consumer price indices



What is provided through scanner data?

Scanner data price information is strictly related to an *article code* and not to a generic product

- Price information derived from scanner data is strictly related to products with common, specific characteristics
- It refers to a specific combination of brand, variety and package with unique identifier code (GTIN, SKU, other). This is known as **article code, elementary item, or individual variety**
- Given that it refers to the monetary transaction, it should include possible discounts on the items



What is provided through scanner data?

An 'average price' is calculated instead, equal to the turnover divided by the quantity, for an article code, in a particular period

“Scanner data sets contain variety quantities sold, and revenue received by the retailer for these varieties for some period of time, usually a week or a month. This information enables NSOs to calculate a price for an individual variety by dividing a variety’s revenue by the quantity sold. This price, referred to as a unit value, represents the average price experienced by consumers over a period of time.”



What do we need for the CPI calculation at the NSO?

individual products listed by day, week or month												package _measur					
												package_typ	package	ement_ unit	unit_pri	sales	quantiti
year	mon	week	outlet	outlet_size	region	product ID	product	Main_category	category_2	category_3	e	_size		ce	value	es	
2020	12	52	55	supermarket	1	6430059782604	Orange juice 0,99 l	Fruits and vegetables	Fruits and berries	Fruitjuice	bottle	0,99	liter	6,5	487,5	75	
2020	12	52	55	supermarket	1	2000614400003	Tomatoes Finnish	Fruits and vegetables	Cucumber and tomatoes	Tomatoes	sale in bulk	-	kg	4,5	270	60	
2020	12	52	68	minimarket	3	6416046151004	Salad 1. class	Fruits and vegetables	Salads and cabbage	Salads	plastic_bag	150	g	2,3	529	230	

Example of essential variables

- **Time** of the purchase (week, month and year)



What do we need for the CPI calculation at the NSO?

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	mon	week	outlet			product						package_typ	package	ement_	ce	value	es
year	th		_ID	outlet_size	region	product ID	description	Main_category	category_2	category_3		e	_size	unit			
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Example of essential variables

- **Time** of the purchase (week, month and year)
- **Outlet type**



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Example of essential variables

- **Time** of the purchase (week, month and year)
- **Outlet type**
- **Retailer** or **outlet region** or **location**



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year	mon	week	outlet			product					package	package	ement	unit	sales	quantiti
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Example of essential variables

- **Time** of the purchase (week, month and year)
- **Outlet type**
- **Retailer** or **outlet region** or **location**
- Product **identification number** (GTIN, SKU, PLU, other), **name** and/or **description**
- Retailer-specific **product categories**



What do we need for the CPI calculation at the NSO?

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- **Product information**: package type, size and measurement unit



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Example of essential variables

- **Time** of the purchase (week, month and year)
- **Outlet type**
- **Retailer** or **outlet region** or **location**
- Product **identification number** (GTIN, SKU, PLU, other), **name** and/or **description**
- Retailer-specific **product categories**
- **Product information**: package type, size and measurement unit
- **Sales value** (turnover/expenditure), **sold quantities**



Requirements for Scanner Data delivery

Key points to note:

- . There is no clear rule regarding whether you should:
 - Define a sample of products and outlets to be included in your scanner data; or
 - Take all scanner data that is available
- . It will depend on:
 - Your plans to introduce scanner data into the CPI compilation
 - Your computing systems and server capacity

Good practice: aggregate transactions by week



An illustration featuring a person's profile in a light teal color, looking down at a document. A dark teal hand holds a pen, pointing at a red 'X' on the document. The document has several horizontal grey bars representing text. To the right is a red folder with a white outline and a small cluster of colorful dots. The background is a solid teal color, and a large orange triangle is on the left side.

3. Negotiation strategy & requirements for scanner data deliveries from large-scale trade retailers

Why is an agreement for scanner data delivery needed?

If CPI compilation is going to be based on the regular acquisition of scanner data from companies, it means that:

- For products involved, the NSO will be abandoning traditional data collection and moving to data collection via a third party
- The NSO no longer has full control of the CPI production process, as some of the data delivery is now in the hands of others
- Any breaks to the supply of data will affect the NSO's ability to deliver the nation's CPI



Why is an agreement for scanner data delivery needed?

It is therefore ***essential*** to negotiate and agree a formal agreement with the data providers

By establishing a formal agreement with the companies, the NSO will:

- Secure a continuous supply of scanner data
- Safeguard against any changes that may risk the future provision of data, such as a change of management at the company



Requirements for scanner data delivery

The NSO's scanner data requirements and terms of the agreement established with the data providers are strictly related



Negotiation strategies and contents of agreement

The Agreement for the scanner data transfer will need to establish:

1. The contents and the characteristics of the data set
2. The rules of the data transmission

Let's take a closer look at these...



Negotiation strategies and contents of agreement

1. The contents and the characteristics of the data set:
 - Detailed description of the elementary items (metadata) & ID code
 - Turnover and quantities sold, by elementary item
 - Data aggregation requirements – weekly or daily transaction data
 - Geographical granularity information (regional variability of prices)
 - **Frequency of delivery of data**
 - **File type and format (csv, txt, xml,...)**
 - **Method of data transfer (SFTP, ...)**



Negotiation strategies and contents of agreement

2. The rules of the data transmission, in terms of:

- Legal elements for data transfer
- Security of the data, data protection, confidentiality & access rights
- Timing and frequency of the data transfer
- Rules for update of the data in case of changes (revisions policy) and in case of unforeseen circumstances (mistakes)
- What service the NSO can provide in return for scanner data
- Duration and termination of the agreement



Negotiation strategies and contents of agreement

- ✓ Request of some preliminary data, for example, request one year of monthly data (best granularity possible) to investigate the weights and prepare the basis of the indices, to:
 - Better determine the granularity available
 - Better define the details within the agreement
 - To prepare the procedures to treat scanner data as far as they will be regularly delivered
 - To perform preliminary studies on methods to be adopted
 - Start construction of IT infrastructure necessary to deal with this source





4. Managing the acquisition of Scanner Data & skills required

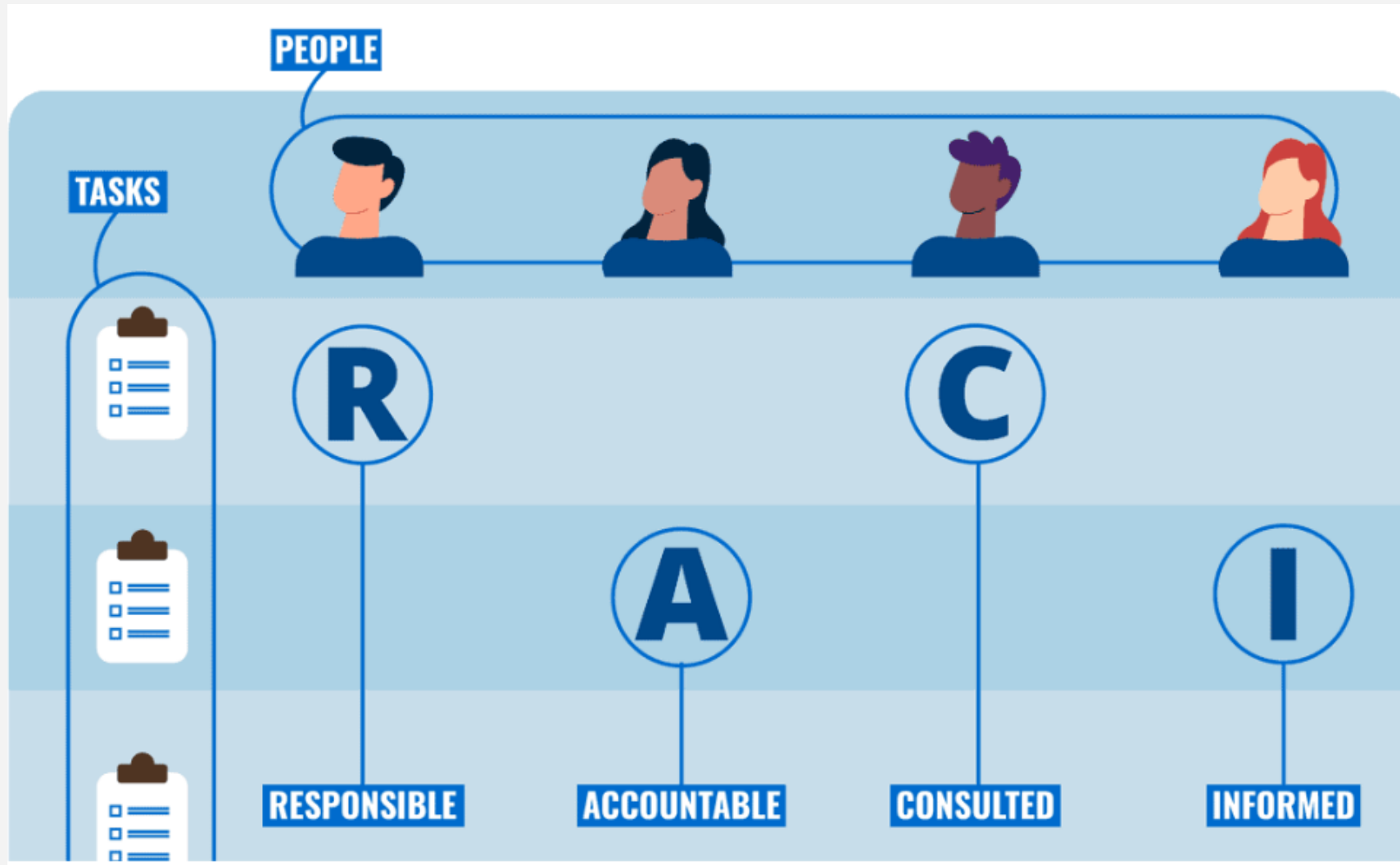


Managing the acquisition of scanner data

- Acquiring scanner data must be managed by the NSO
- The NSO must identify different stakeholders and assign responsibilities – like managing a project
- The NSO must identify different issues and deal with them accordingly
- A RACI matrix is a good tool to support the acquisition of scanner data at the NSO

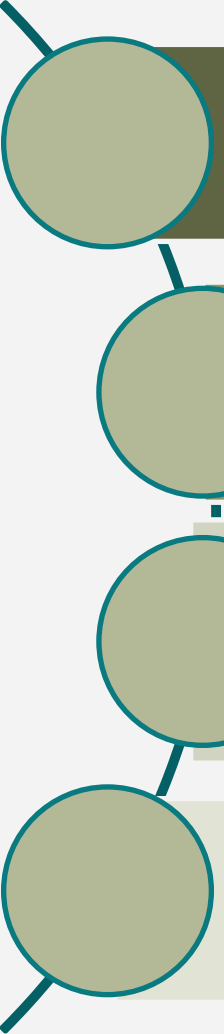


What is a RACI matrix?



What is a RACI matrix?

Project roles required



Responsible. Carries out the work to complete the task through to delivery. Each task should have at least one Responsible person (or more) and should be in the project team.

Accountable. Reviews the work involved and makes sure the responsible person/people complete their tasks on time. There should be only one Accountable person in this project leadership or management role.

Consulted. Contributes before and throughout the work, providing useful information for the project and participating in two-way communication. This person may be internal or external to the project team, but they will be affected by the outcome of the project.

Informed. Parties and persons interested who must be kept informed of the state of play of the project. They should be external to the project team - they may be Heads of affected teams or senior leadership in the organization.

The RACI matrix: Acquiring Scanner Data

Tasks	Economists	Statisticians	Project Manager	Data Analysts	Data Scientists	IT experts	Retailers
Analysis of the market	R	R	A	I	I	I	C
Analysis of the level of concentration/fragmentation of the large-scale retail trade market	R	R	A	I	I	I	C
Promote bilateral contacts and meetings with companies/national branches of GS1/Associations of companies	C	C	A/R	I	I	I	C
Start negotiations with specific companies/national branches of GS1/Associations of companies	C	C	A/R	I	I	I	C
Specify scanner data requirements	C	R	A	C	C	C	C
Finalize with agreement and signatures	I	C	A/R	I	I	I	C
Data treatment and data cleaning	I	C	A	R	R	R	





5. IT infrastructure and monitoring



IT infrastructure



STORAGE SPACE



**COMPUTING
POWER & RAM**



PEOPLE



Monitoring

- Once established, constant monitoring of data received from retailers is necessary.
- Manual inspection of the whole data set is not possible, creation of automatic checks via quality indicators is necessary.
- Checks should be performed at data files at entrance.



Monitoring

Checks can include:

- Evaluating if all the expected variables in the proper formats are received
- Changes in file sizes
- The number of new and disappearing item elements
- Turnover and quantities for a given elementary item is comparable over time
- If data is being provided in the timetable established
- Check for null or negative prices and turnover



6. Highlights



Highlights

- . We have seen how to develop a strategy to identify suitable sectors for using scanner data and suitable retailers and optimize negotiations
- Identified main variables of interest for a scanner data set
- Discussed necessity and main contents of an agreement for obtaining scanner data
- Presented discussion on how to organize a project for data acquisition according different skills and actors involved.
- Highlighted importance to build a robust IT infrastructure and perform constant monitoring of the data received to avoid problems and maintain the quality of the CPIs produced.





2. Scraping data from the web

Contents

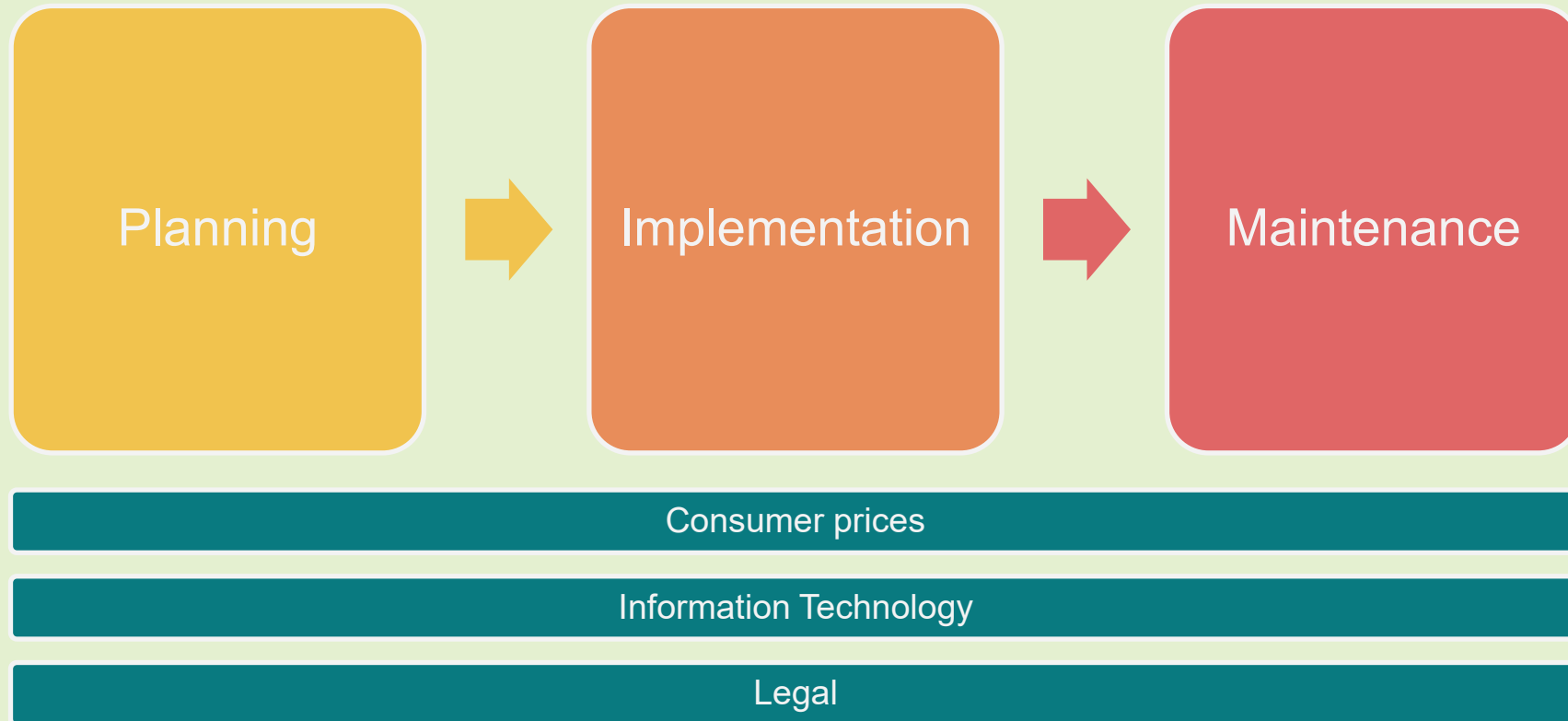
1. The web scraping process
2. Planning
3. Web scraping implementation
4. Highlights



1. The web scraping process



Web scraping data acquisition





2. Planning



Environment investigation

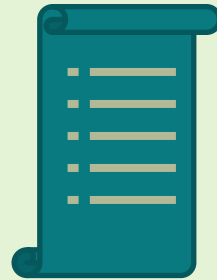


Define target categories

Consider specific complexity



Data model



Define a consistent data model

For instance:

- Product name
- Brand
- Size
- Unit of measure
- Price



Scraping volume approach



Targeted



Bulk



Scraping volume approach: examples



Air fares:

- Selected routes
- Selected dates
- Selected carriers

Targeted



Clothing:

- Selected brand websites
- All products
- Sample selected post scraping

Bulk



Law and etiquette



Legal framework

Reputational issues

Example: Statistics Finland has published its general operating principles concerning web scraping. You can retrieve them via the link in the slide for your reference.



Relevance

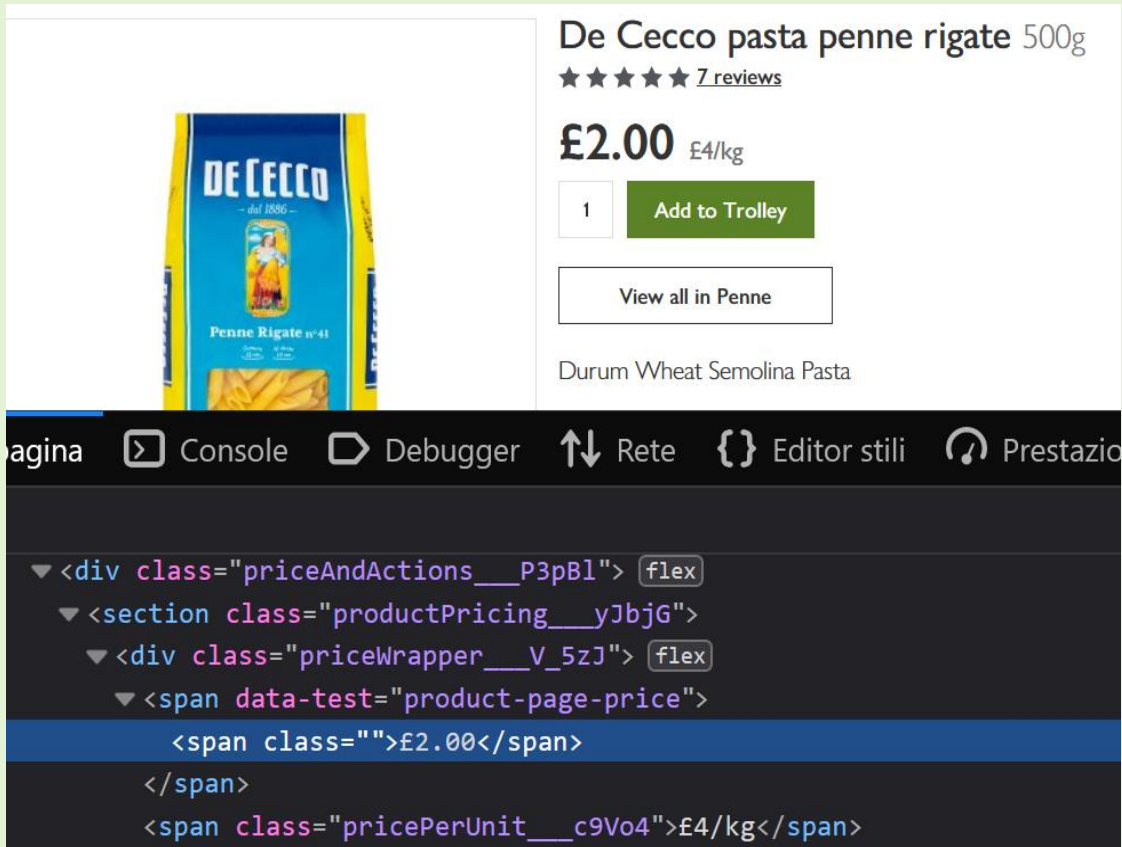


Purchase habits: determine how relevant are those prices for the population in your country

Multichannel retailers: explore whether online and offline retailers are the same, and whether they practice the same prices across sales channels.



Website analysis



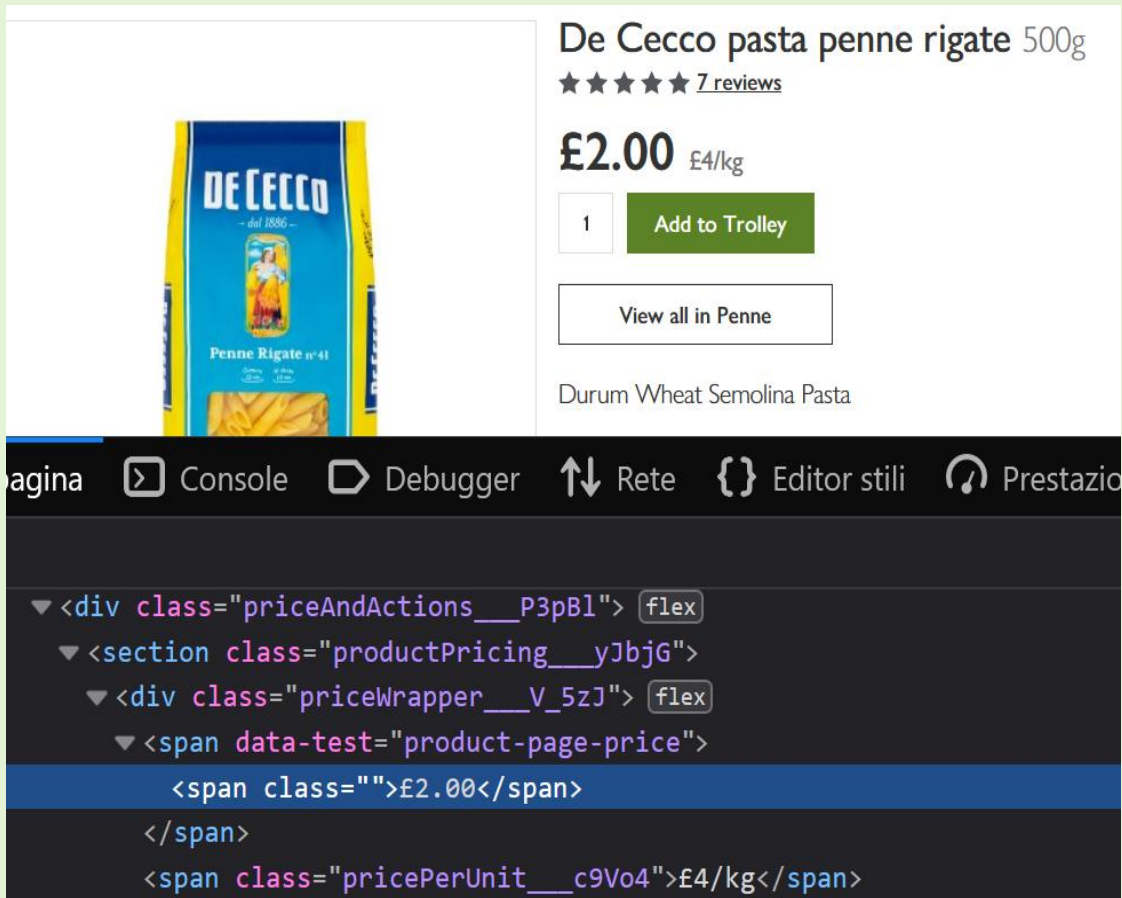
The screenshot shows a product page for De Cecco pasta. The product is 'De Cecco pasta penne rigate 500g', rated with 5 stars and 7 reviews. The price is £2.00 (£4/kg). There is an 'Add to Trolley' button and a 'View all in Penne' button. The product is described as 'Durum Wheat Semolina Pasta'. Below the product image, a browser developer console is open, showing the HTML structure. The console highlights the following code:

```
<div class="priceAndActions__P3pB1"> flex
  <section class="productPricing__yJbjG">
    <div class="priceWrapper__V_5zJ"> flex
      <span data-test="product-page-price">
        <span class="">£2.00</span>
      </span>
      <span class="pricePerUnit__c9Vo4">£4/kg</span>
```

Once you have defined a list of target websites, their specific structure. One of the first elements to look for is the robots.txt file. The information available on this file is not standard, some website may even not have one. In most cases, the **robot.txt** file includes information about the website structure and what operations the website operator is allowing or disallowing on specific webpages. While the content of the robots.txt file is not legally enforceable, it is considered proper behavior to respect it



Website analysis



The screenshot shows a product page for De Cecco pasta. The product is 'De Cecco pasta penne rigate 500g', priced at £2.00 (£4/kg). The page includes a star rating of 5 stars with 7 reviews, an 'Add to Trolley' button, and a 'View all in Penne' button. The developer tools are open, showing the HTML structure of the price and actions section. The selected element is a with class="" and text content £2.00.

De Cecco pasta penne rigate 500g
★★★★★ 7 reviews
£2.00 £4/kg
1 Add to Trolley
View all in Penne
Durum Wheat Semolina Pasta

agina Console Debugger Rete Editor stili Prestazio

```
<div class="priceAndActions__P3pB1"> flex
  <section class="productPricing__yJbjG">
    <div class="priceWrapper__V_5zJ"> flex
      <span data-test="product-page-price">
        <span class="">£2.00</span>
      </span>
      <span class="pricePerUnit__c9Vo4">£4/kg</span>
```

Websites may be static or dynamic, and some may also use anti-scraping tools. You can use the Developer Tools from any modern web browser to explore the website and define the most appropriate web scraping methodology. You should also consider registering your web scraping software (called “bot”, “spider”, “crawler”, “or “scraper” in web scraping jargon) with the main platforms that provide anti-scraping services, such as Cloudflare, in order to avoid being blocked.



Contact



As a final part exploration, consider contacting the website operators explaining your intention to access their websites via web scraping, and explaining your purpose for the use of such data.

You may agree with them specific times and modalities for data collection that may better suit them. In some cases, they may even offer you alternative ways to access the data you need.

Those alternative ways in general are preferred, since they may provide additional information (such as quantities sold), and more stable





3. Web scraping implementation



Before production



Before including any new data into your official CPI process, you should undergo a pre-production plan and monitor the CPI from web scraping for a reasonable period, indicatively 6 to 18 months.

Aggregation weights and integration structure may require fine tuning to avoid disruption in the overall CPI index.



Contingency planning



Before including web-scraping data into the official CPI process prepare a contingency in case web scraping is disrupted at any point in the future. Short-term breaks are not a major concern, whereas long-term or permanent breaks require radical interventions in the aggregation structure and weights to avoid distortions.

Another point to keep is continuous monitoring. Since the volume of data collected via web scraping may be huge, manual control is not an option. Design an automated quality control pipeline.





4. Highlights



Phases of acquiring and using web-scraped data

- Start from planning
- Expertise from multiple areas needed
- Build data pipelines
- Update pipelines over time
- Pipeline monitoring and contingency plans
- Production





3. The APIs data acquisition process

In your strategic planning you should define some key points, such as which type of products and services to include in your data collection and the data collection frequency.



API providers evaluation

Coverage

Rate
limits

Historical
data

Update
frequency

Reliability

Pricing



Data pipelines implementation



After selecting the API providers for your data acquisition, you can move on to implementing the data acquisition and classification pipelines. This is relatively straightforward compared to other data sources, as the API provider usually produces documentation about the API use, and many of them may also be able to provide technical support.



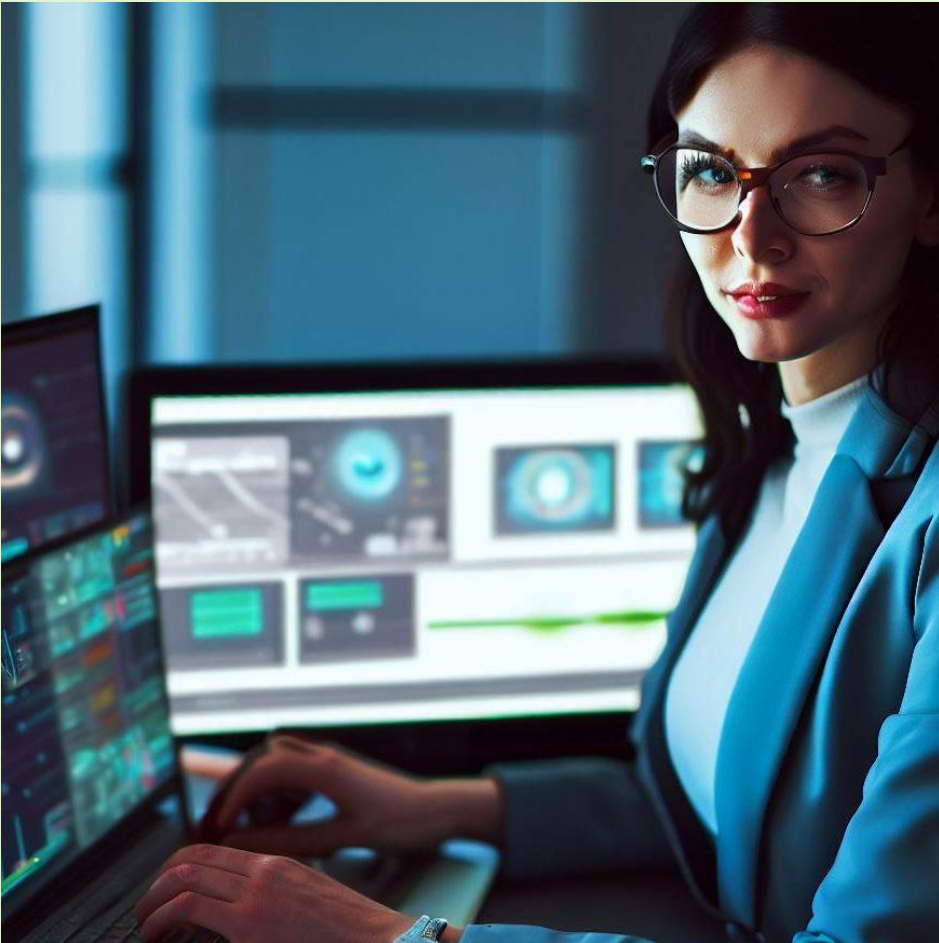
Data pipelines implementation



Define in detail your queries to create a stable basket of products and services you monitor over time. Some practical suggestions are:

- Keep the price statisticians in control of the parameters for the queries, and provide them an interface to check and modify those parameters (for airfare prices: departure city, destination, number of days before flight, travel class. Each combination of those parameters corresponds to a different service
- Data from API is generally in a consistent format. However, it is recommended to perform quality checks before inserting them into your database. Most API providers will issue a notice in case something changes in their API, and it is good practice for them to use versioning. This means that if they release a new version of their API, it will have a different endpoint for your API calls

Ongoing maintenance



The next thing that you should consider when including API data into the official CPI process is to perform periodical reviews for the available API providers and their price structure.

Changes in the price structure may substantially alter the budget for your data collection. Like any other source of data, you should assess and compare its cost efficiency.





4. Administrative Data Acquisition

Contents

1. Motivation of administrative data for CPI compilation
2. Planning for Administrative Data Acquisition for CPI
3. Interaction with Authorities for Administrative Data
4. Resources and Skills for Administrative Data Acquisition
5. Case Studies and Examples
6. Conclusions and Final Assessment



Motivation of administrative data for CPI compilation

Motivation of administrative data for CPI compilation

- The digitalization has, amongst the others, the effect of increasing the production of administrative records and producing data sets within the scope of price statistics.
- To manage the tasks of the government and public authorities, digital systems and procedures have been developed to accomplish these tasks more efficiently and facilitating relations with families and businesses (for instance income recording or electronic invoice or rentals contracts registration for taxation aims)
- Thus, several data sets are produced for administrative aims and a lot of these data sets can be used for statistical aims
- Here we will discuss the main issues about the processes and characteristics of the acquisition of administrative records for CPI compilation.



Planning for Administrative Data Acquisition for CPI

Planning for acquisition of administrative data for CPI

- Availability of sources for price statistics in administrative records may vary for different countries as well as possible basket sectors covered, and the characteristics of the variables contained in such data.
- Thus, the first step to start a plan of administrative data acquisition is to map to the administrative data sources potentially available to the aggregates of COICOP (or other households' consumption classification adopted by the country)
- In the following table an example is given about how to map all the administrative data sources to the COICOP 2018 Divisions, identifying their potential coverage for consumer price indices compilation aims



Planning for acquisition of administrative data for CPI

CO DE	COICOP 2018 divisions	Large scale retail trade channel – scanner data	Traditional distribution – direct data collection	Service providers - direct data collection	Service providers – administrative data	Internet purchase – web traditional data collection	Internet purchase – web scraping	Other distributional channels – other sources	TOTAL
01	FOOD AND NON-ALCOHOLIC BEVERAGES	60%	20%	0	0	20%	0	0	100%
02	ALCOHOLIC BEVERAGES, TOBACCO AND NARCOTICS	50%	10%	0	40%	0	0	0	100%
03	CLOTHING AND FOOTWEAR								100%
04	HOUSING, WATER, ELECTRICITY, GAS AND OTHER FUELS	0	0	50%	30%	10%	10%		100%
05	FURNISHINGS, HOUSEHOLD EQUIPMENT AND ROUTINE HOUSEHOLD MAINTENANCE	20%	30%	10%	0	20%	10%	10%	100%
06	HEALTH	0	0	40%	30%	10%	10%	10%	100%
07	TRANSPORT	10%	50%	20%	20%	10%	0	0	100%
08	INFORMATION AND COMMUNICATION	20%	10%	20%	10%	20%	20%	0	100%
09	RECREATION, SPORT AND CULTURE	20%	40%	5%	0	10%	10%	15%	100%
10	EDUCATION SERVICES	0	20%	20%	60%	0	0	0	1005
11	RESTAURANTS AND ACCOMMODATION SERVICES	0	40%	30%	0	20%	10%	0	100%
12	INSURANCE AND FINANCIAL SERVICES	0	10%	50%	20%	20%	0	0	100%
13	PERSONAL CARE, SOCIAL PROTECTION AND MISCELLANEOUS GOODS AND SERVICES	20%	20%	20%	20%	10%	0	10%	100%

Planning for acquisition of administrative data for CPI

- In the previous table a general map of the potential sources to acquire data for CPI compilation is sketched and its usefulness goes beyond the topic of administrative data acquisition
- In the table, for 8 COICOP 2018 divisions administrative data sources are potentially identified. A key step related to this is then the identification of potential sectors of the government that oversee such information. This kind of evaluation is country-dependent, but usually potential candidates are tax authorities, ministries and regulatory agencies which usually have detailed data mainly for fiscal aims about a given sector.



Planning for acquisition of administrative data for CPI

- As a matter of example:
 - For Division 2, tobacco products are often under the government monopoly and the related system of taxation is strictly regulated and this means that price data are regularly available at the charged Ministry
 - For Division 4, data sets containing the information about the rental contract are available at the Tax Authorities and potentially usable for CPI compilation aims
 - For Division 10, information about the number of students and fees paid for private schools or University are collected at the Ministry of Education



Planning for acquisition of administrative data for CPI

- After the **mapping**, source by source it is useful to extract further details on the **data owner** (tax authorities, regulatory agency, ministry, others) and on the **contents** of each source: what are the variables, what is the coverage in terms of each product of the basket, the geographical breakdown of the data and market coverage, the possible presence of **personal data** etc.
- How to handle and to protect personal data has to be setup ahead of starting the process of acquisition, by considering the legislation in force in the country



Planning for acquisition of administrative data for CPI

- Beyond the contents, it is needed to evaluate how such kind of data could be acquired by the NSO from the other governmental sectors and to this aim other two key issues have to be considered in the planning phase:
 - Acquisition can be facilitated by the **statistical law** of the country in case it promotes that administrative records should be shared with the NSO to produce statistics. However, even with a law at hand negotiations might be required to address possible issues and practical agreements for the data transfer process
 - If no statistical law is in force, or the statistical law does not have specific disposals about access to administrative data sets for statistical aims, exploring if there is **any kind of agreement** that empowers the NSO to get such data or even if the data are already available on demand is a starting point



Interaction with Authorities for Administrative Data

Interaction with Authorities for administrative data

- Negotiations with Authorities and Entities at the beginning of the process of acquisition of administrative data for CPI aims are needed to tailorize the provision
- Administrative data sets are built up by administrative authorities for aims other than the statistical ones and their usage for these aims requires some specific precautions in terms of variables to be transmitted and in terms of frequency, timing, and modalities of transmission



Interaction with Authorities for administrative data

- Concerning the contents, the National Statistical Office should send to the administrative Authority, in charge of collecting the data for a certain specific domain, **the list of variables** needed for a correct CPI compilation for that specific product
- Together with the list of variables, it is recommendable to specify the other needs related to:
 - The **frequency of transmission** (monthly in principle), except for some products as Education services (fees for University and school change once a year but, anyway, a monthly monitoring is recommendable)
 - The **timing** of transmission
 - The **record track and transmission protocol**;



Interaction with Authorities for administrative data

- One of the most critical points in the negotiations with the administrative Authorities is related to the **timing** of transmission:
 - The CPI timing and the timing of administrative Authorities could not coincide given that CPI is mainly produced on monthly basis
 - It is crucial to establish specific and precise rules in terms of timing of data transmission to the National Statistical Office
- The outcome of the negotiations should be an **agreement** (formal if there are no pre-existing agreements or a statistical law that promotes the sharing administrative data with Statistical Authorities) that establishes **(i)** variables, **(ii)** frequency, **(iii)** timing, **(iv)** record track, **(v)** transmission protocol and **(vi)** governance rules of data provision



Resources and Skills for Administrative Data Acquisition

Resources and skills for administrative data acquisition

- In the case also of administrative data acquisition, a RACI matrix provides a simple but effective way of defining project roles and responsibilities at the National Statistical Office and, consequently, an overview of the resources and skills that are needed
- RACI (for further explanation, please consult Module 2.1) is the acronym for:
 - Responsible
 - Accountable
 - Consulted
 - Informed



Resources and skills for administrative data acquisition

Tasks	Economist	Statistician	Legal expert	Project Manager	Data Analyst	Data Scientist	IT expert	Administrative Authority
Mapping potential admin data sources to COICOP aggregates	R	R	-	A	I	I	I	-
Analysis of the contents of Administrative data sets	R	R	I	A	I	I	I	C
Managing personal data protection issues	C	C	R	A/R	I	I	I	C
Start negotiations with Authorities in charge of collecting information for administrative aims	C	C	C	A/R	I	I	I	C
Specify administrative data requirements	R	R	C	A/R	C	C	C	C
Finalize with agreement and possible signatures	I	C	C	A/R	I	I	I	C
Data treatment and data cleaning	I	C	C	A	R	R	R	-



Case Studies and Examples

Case Studies and Examples

- There are several examples around the world of successful acquisition of administrative data for CPI compilation aims
- Considering the references used in module 1.4, one of these cases is that of **Italian National Statistical Institute (Istat)** and of the acquisition of **rentals for housing data from Italian Tax Authority** (concerning when the rentals for housing data sets managed by Tax Authority became suitable for CPI aims and their contents, please, look at module 1.4)
- Here, it is interesting to learn about the process of acquisition and the negotiations that took place



Case Studies and Examples

- The acquisition process and the negotiations developed through the following milestones:
 1. Establishing a technical working group with the Observatory of Real Estate Market (about the provision of information on real estate domain) of the Italian Tax Office under the umbrella of the Memorandum of Understanding between Istat and the Italian Tax Office (2011) and the more general statistical Law
 2. After the discussion and the acquisition of data about purchase prices of dwellings, a discussion started in 2017 and 2018 about acquisition of rentals for housing data sets as soon as the digitalization of the registration of contracts of rents was completed and harmonized
 3. Istat specified statistical requirements for CPI compilation



Case Studies and Examples

4. The most important issues emerged about the frequency (quarterly availability of data, versus monthly requirement) and the timing (quarter t data available in the middle of quarter $t+1$). Istat asked for increasing the frequency (from quarterly to monthly elaboration of data) and reducing the lag of data release, giving evidences of the potential improvements in the estimation of inflation of this component of the basket, deriving from the use of such a complete source of data.
5. Successful work of the Observatory of real estate market with Tax Office statistical unit in terms of frequency of data elaboration (brought from quarterly to monthly) and to sort out the issue of timing (a compromise was achieved, accepting one month of lag compared to the nearly census monthly data)



Case Studies and Examples

6. Definition of the record track, anonymization of micro data sets of rentals for housing and test of the transmission protocol, following the same used for the house prices
7. About one year (2021) of test ahead of adopting the use of the new nearly census data for official CPI compilation (since January 2022)



Conclusions and Final Assessment

Conclusions and final assessment

Here the main takeaways from this module on administrative data acquisition:

- **Understanding Administrative Data:** the module emphasizes the importance of administrative data as a valuable source for consumer price statistics compilation. It highlights the main issues and challenges associated with acquiring such data
- **Planning for Administrative Data Acquisition:** It is crucial to map potential administrative data sources to COICOP aggregates. This involves identifying the coverage and exploring which governmental sectors produce or demand the data of interest for CPI



Conclusions and final assessment

- **Negotiations with Authorities:** Effective negotiations with authorities are essential to tailor data provision and establish transmission protocols. This includes specifying variables, frequency, timing, and modalities of data transmission
- **Resources and Skills:** A RACI matrix is recommended to define roles and responsibilities for managing administrative data acquisition. This ensures that the necessary resources, knowledge, and skills are in place
- **Case Studies:** The presentation provides examples, such as the Italian case, to illustrate the administrative data acquisition process for CPI aims





5. Conclusions

Conclusions

- Brief overview of general concepts
- Main similarities and differences in data acquisition for different alternative data sources.



Conclusions - general

- The process of acquiring alternative data sources is completely different compared to traditional price data collection.
- In all cases of alternative data acquisition, NSOs no longer have complete control of the data-acquisition process.
- Acquisition of alternative data usually requires an agreement with data providers (but the type of agreement can differ between ADS types).



Conclusions - common characteristics

- Data are not originally designed or structured for statistical use.
- Improved CPI coverage of products, time and geographical regions makes more complex the data acquisition.
- Negotiations and agreements with data providers are necessary.
- Reduced respondent burden and, in most of the cases, increased timeliness of data acquisition.
- Data can be acquired by a more controlled and centralized.
- Specific IT tools are necessary to acquire, store and process the new data received
- Periodic monitoring of data quality, providers and costs is required.
- Pre-production testing and developing contingency plans is strongly recommended.



Conclusions – differing characteristics

- Data contents can differ (scanner data brings information on quantity, geographical breakdown can be differently detailed) and this could imply setting up different process for data acquisition.
- Scanner data and administrative data (in some cases also APIs) require actions from data "owners" to send data to the NSO:
 - ❖ File transfer process and storage must be created.
 - ❖ More detailed negotiations is required (for instance about timing).
- Whereas web-scraped and data from APIs (if not specific requirements are necessary) data do not need additional contributions from the data "owners".
- Different skills are required for different alternative data sources, and the RACI matrix has to be well designed considering the specificities of each of them.

